ATTACHMENT IV. JOINT IMF-WORLD BANK DEBT SUSTAINABILITY ANALYSIS

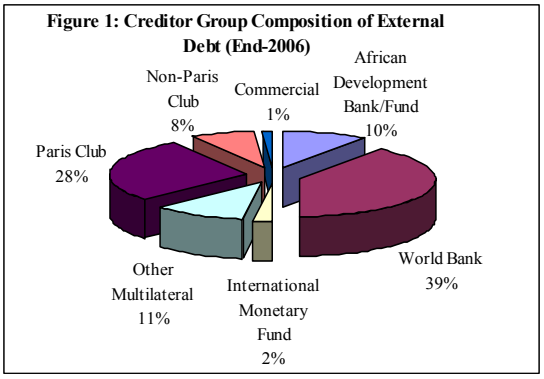
***Guinea is in debt distress**, and would remain at a high risk of debt distress into the medium term in the absence of the country's reaching the Completion Point under the Enhanced HIPC Initiative and benefiting from the Multilateral Debt Reduction Initiative (MDRI). If the HIPC Completion Point is reached by end-2008, the external debt burden will decline to a sustainable level from that point. Results of the public sector DSA indicate that Guinea's domestic debt is significant but is expected to decrease over time.*[1]

A. Background

1. **This debt sustainability analysis (DSA) updates the analysis of the external and public debt of Guinea that was carried out in 2005**.[2] It applies the methodology of the Fund-World Bank debt sustainability framework (DSF) for low income countries. In accordance with the framework, this DSA compares the expected trajectory of debt indicators with the thresholds for countries with weak policies and institutions (Table 1).[3]

2. **Guinea reached the decision point under the Enhanced HIPC Initiative in December 2000, qualifying for US$545 million (in NPV terms) in debt relief.** In 2001–02, interim debt relief was provided by all main creditors including the IMF, the International Development Association (IDA), the African Development Bank (AfDB), and Paris Club members. After the PRGF went off-track at end-2002, many creditors, including the majority of Paris Club members, discontinued interim relief. At the same time, the AfDB continued interim relief but eventually exhausted resources allocated to that purpose. Only IDA and two Paris Club creditors continued providing interim relief to date. As a result of policy slippages and dwindling debt relief, Guinea has had difficulties servicing its external debt since 2003.

3. **At end-2006, Guinea's public and publicly guaranteed external debt was US$3,188 million, or 101.5 percent of GDP**. Multilateral creditors accounted for 63 percent of the total, with the IDA and the AfDB group accounting for more than four-fifths of the multilaterals' share (Figure 1). The Paris Club group of creditors accounted for 28 percent of total, while non-Paris Club and commercial creditors made up the rest. Like many HIPC countries, the

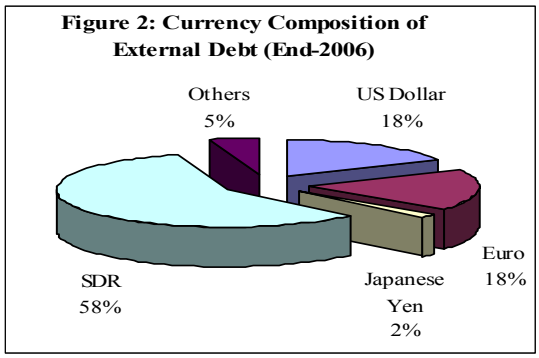


[1] The DSA has been produced jointly by Bank and Fund staffs. The fiscal year for Guinea is January 1 to December 31.

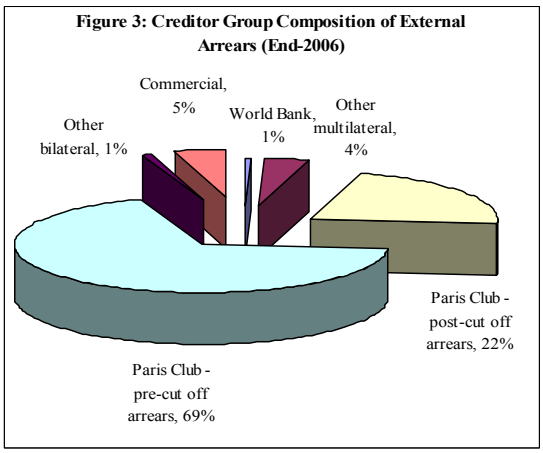
[2] Appendix V of IMF Country Report No. 06/37, January 2006.

[3] Guinea's average score under the World Bank's Country Policy and Institutional Assessment (CPIA) in 2004-06 was 2.9.

currency composition of Guinea's external debt leaves it vulnerable to changes in the U.S. dollar exchange rate against major international currencies. With 78 percent of the external debt denominated in Euro, Yen and SDR (Figure 2), the depreciation of the U.S. dollar against major international currencies since the decision point has increased the U.S. dollar value of the debt.



4. **The external debt service arrears reached nearly US$191 million at end-2006** (Figure 3). The Paris Club members accounted for 90 percent of the arrears, and commercial creditors and multilateral institutions for 5 percent each, including 1 percent for IDA (there was no overdue obligation to the IMF).[4] Arrears to the Paris Club members accrued on both pre-cut-off and post-cut-off debts. The former largely represents arrears on payments covered by the interim HIPC assistance that was discontinued by a majority of creditors after the PRGF program went off-track at end-2002.



B. External Debt Sustainability Analysis

Baseline scenario

5. The baseline macroeconomic framework underpinning this DSA assumes implementation of sound macroeconomic policies and structural reforms, focusing on those aimed at improving governance. The average growth assumption (5 percent during 2008–27) is also predicated on measures to remove bottlenecks to the business climate, invest in infrastructure, and promote the provision of electricity and other basic services. In addition, the implementation of large mining projects is expected to lead to the gradual expansion of mining production from 2010 onwards (Box 1).[5]

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[4] Arrears are defined as payments past due date. This may differ from the definitions applied by the creditors.

[5] The baseline is consistent with the macroeconomic framework envisaged for the period 2007–10 in the context of a program to be supported by a PRGF arrangement and the 2007 consultation under the Article IV of the IMF articles of agreement. However, the latter reflects the impact of the debt stock reductions under the enhanced HIPC and MDRI initiatives that would result from Guinea reaching the HIPC completion point. The HIPC completion point and MDRI debt relief is presented here in the alternative scenario.

6. **Consistent with current guidelines,[6] the baseline scenario assumes only interim flow relief in 2008.** It is projected that some creditors will provide interim HIPC debt relief according to the HIPC decision point and the subsequent Paris Club debt agreement.

7. **Under the baseline scenario, Guinea is in debt distress** (Table 1, Figure 4)**.** At end-2006, the debt burden indicators were well above the indicative policy-based thresholds. Also, the country has accumulated substantial external arrears in the recent years. In the baseline scenario, several debt burden indicators are projected to remain above the thresholds for the next 10–12 years.[7] In particular, the NPV of debt-to-exports ratio is estimated at 186 percent at end-2006, well above the indicative threshold of 100 percent, and is not projected to go below this threshold until 2020. However, the ratios of debt service to export and revenues can fall below the thresholds as soon as in 2008 reflecting both the projected resumption in interim HIPC debt relief by some creditors and a projected increase in fiscal revenues (Box 1).

Table 1: Policy-Based External Debt Burden Indicators

	Thresholds [1]	Guinea's Ratios							
		Baseline					HIPC and MDRI scenario		
		2006	**2007**	**2008**	**2009**	**2008-27 [2]**	**2008**	**2009**	**2008-27 [2]**
NPV of debt in percent of:									
Exports	100	186	160	157	146	107	59	60	67
GDP	30	65	41	43	41	29	16	17	18
Revenues [3]	200	530	364	313	282	187	119	115	115
Debt service in percent of:									
Exports	15	19	16	8	13	8	8	5	5
Revenues [3]	25	53	37	15	24	14	15	10	8

[1] Policy indicative thresholds as used in the joint IMF-World Bank LIC DSA framework for a weak policy performance. The quality of policies and institutions is measured by the World Bank's CPIA.
[2] Simple average.
[3] Revenues excluding grants.

[6] International Monetary Fund, 2007, "Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries," IMF Policy Paper and IDA document no 39748.

[7] The thresholds are based on the empirical finding that low-income countries with stronger policies and institutions tend to have a higher debt carrying capacity. See IDA and IMF, "Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations" (IDA/R2005–0056), April 2005.

Box 1. Guinea: Macroeconomic Assumptions Underlying the DSA

Real GDP growth is projected to pick up from the average of 3½ percent in the past ten years to an annual average of 5 percent over the projection horizon. The higher growth is predicated on government's planned policies to consolidate macroeconomic stability, upgrade and expand public infrastructure, and strengthen governance. Also, several large mining projects are currently in the implementation stage; they are expected to boost production capacity from 2010 onwards.

Average annual **inflation** as measured by the CPI is expected to decelerate sharply from 35 percent in 2006 and an expected 23 percent in 2007 to single-digit levels in 2009 and to stabilize at 5 percent from 2011 onwards. This inflation projection reflects the authorities' commitment to refraining from monetary financing of the budget deficit and strengthening the central bank capacity to use policy instruments for controlling monetary aggregates.

The **GDP deflator** is expected to be broadly in line with the CPI inflation, while the GDP deflator in U.S. dollar terms will take into account the US inflation, projected to remain within 2–3 percent. This implies a stabilization of the real exchange rate from 2008. For 2007, however, the baseline scenario envisages a strong increase of the nominal GDP in U.S. dollar terms consistent with the appreciation of the local currency in the first ten months of the year.

Export growth is projected to average at about 7 percent over 2007–10 and about 8 percent over 2011–27. The medium-term export projections are affected by the large investments in the mining sector. Excluding official transfers, the **external current account deficit** is assumed to expand from 6.5 percent of GDP in 2006 to 15.2 percent in 2011 and then gradually decrease to 13 percent by 2027, reflecting the projected increase in FDI-related imports over this period.

The **primary fiscal surplus** (including grants) is projected to stay positive over the medium term, gradually decreasing from 2.4 percent of GDP in 2007 to 1.2 percent in 2010 and to 0.7 percent on average in 2011–27. The envisaged broadening of the tax base and strengthening of customs and tax administration would help raise revenue by about 3 percentage points of GDP over the next five years. At the same time, public expenditures would have to increase to address the substantial social and infrastructure needs of the country.

Donor support is projected to increase, assuming the timely implementation of the authorities' new poverty reduction strategy. New grants and concessional loans are expected to be mobilized in equal proportions on top of the already committed financing, which is biased towards loans. For the new loans, the average grant element is projected to remain significantly above the standard threshold of 35 percent for the time of the possible new PRGF arrangement (2007–10), and would slowly decline thereafter.

8. **The baseline macroeconomic projections are subject to significant risks.** A relaxation of fiscal and monetary policies and protracted delays in the strengthening of the central bank and public finances or in the implementation of an effective regulatory and attractive tax frameworks for key sectors of the economy would put into question the assumptions on growth, external support and official reserve accumulation. They would also increase Guinea's external vulnerability.

9. **Guinea's debt burden indicators are particularly sensitive to exogenous shocks and a reversal to historical trends.** Sensitivity analyses indicate that the baseline debt burden indicators are most vulnerable to exogenous shocks to export growth and to a one-time depreciation of the Guinean franc against the U.S. dollar. Under these most extreme stress tests, debt ratios exhibit a significant level increase and are slower to return to policy-based thresholds (see the most extreme stress tests in Figure 4). Sensitivity to exchange rate assumptions is also illustrated in a scenario where key macroeconomic variables remain at their historical averages from 2007 onwards. Under this scenario, Guinea's external debt ratios would snowball, reflecting mostly the assumed continuation of the sharp real exchange rate depreciation that Guinea experienced over the past ten years.

Alternative scenario envisaging HIPC completion point and MDRI

10. **An alternative scenario that simulates HIPC and MDRI debt relief indicates a sharp reduction in the risk of debt distress at the expected HIPC completion point** (Figure 4, HIPC/MDRI scenario). The HIPC debt relief would reduce projected annual debt service due to Paris Club creditors by about three quarters.[8] Upon reaching the HIPC completion point, the country will also become eligible for debt relief under the MDRI, which would write-off all debt obligations to IDA incurred up to end-2003, and the IMF and AfDB debt obligations incurred up to end-2004.[9] The implementation of HIPC and MDRI debt relief at end-2008 would eliminate about 60 percent of Guinea's outstanding external debt at end-2008 (in net present value terms). This would proportionally reduce Guinea's external debt ratios at end-2008, which would remain well below the policy-based thresholds afterwards (Table 1).

11. **Even after the completion point is reached, Guinea's debt sustainability will remain vulnerable to adverse exogenous shocks—especially from exchange rate fluctuations—and policy reversal.** This highlights the importance of rapid achievement of the Completion Point, sustained implementation of sound fiscal, monetary and structural policies, and a prudent debt strategy, which includes contracting external loans only on concessional terms.

C. Public Sector Debt Sustainability Analysis

12. **Guinea's domestic debt is significant but is expected to decrease over time.** At end-2006, identified government domestic debt amounted to about 15 percent of GDP, of which about three quarters consisted of advances from the central bank, the rest being treasury bills owned by commercial banks (6 percent) and securitized arrears to the private sector (20 percent). The baseline macroeconomic scenario assumes a reduction in the level of domestic debt, as a result of the projected fiscal consolidation and elimination of central bank financing of the budget.

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[8] The staff estimates for HIPC completion point debt relief are based on authorities' data and subject to loan data verification and confirmation in the detailed pre-completion point debt analysis (HIPC DSA).

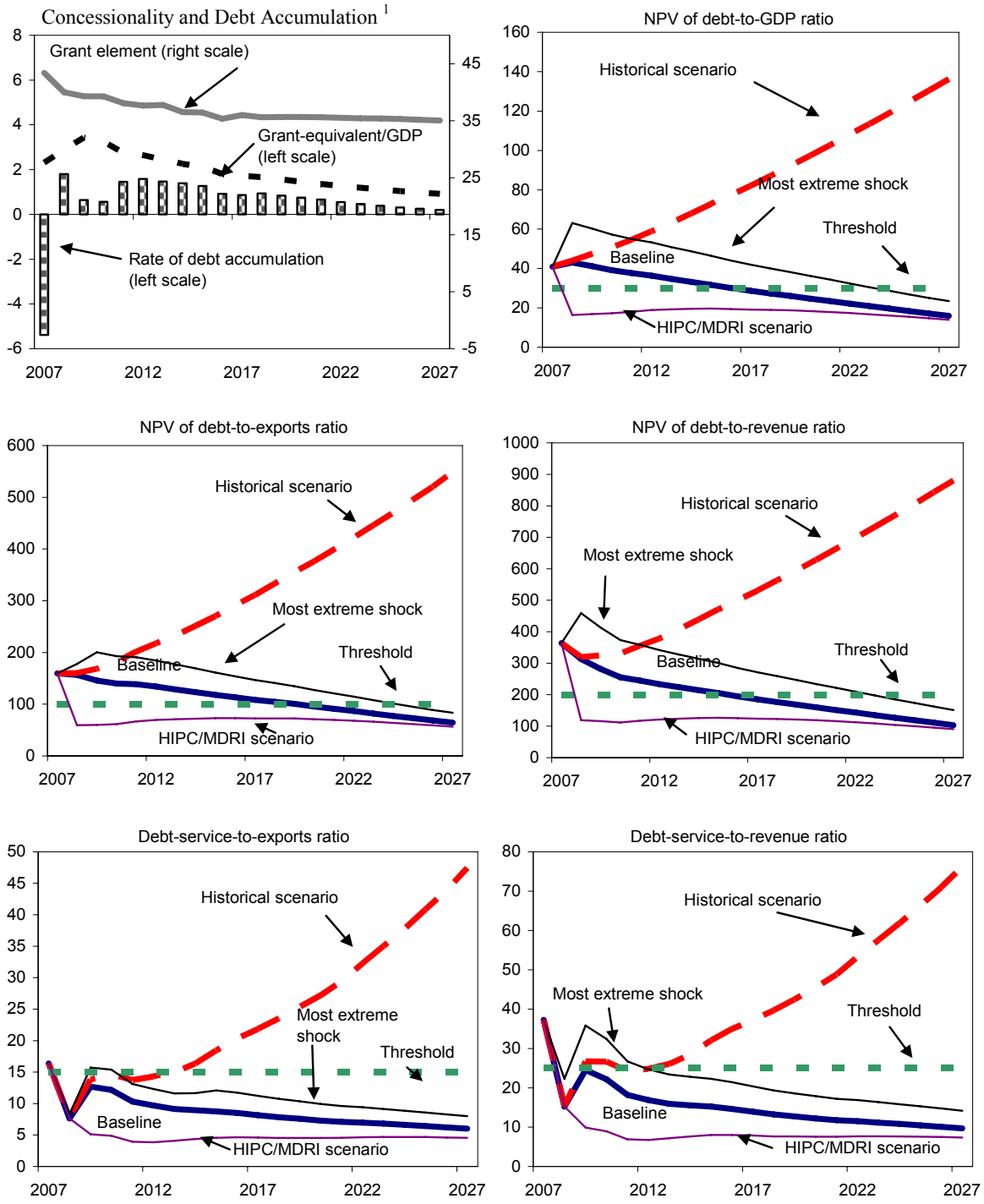
[9] The MDRI does not impact the status of new debt accumulated to multilateral creditors after those dates.

13. **The debt sustainability analysis shows a relatively rapid improvement of public sector debt indicators** (Table 4). The NPV of public sector debt is projected to decline from over 80 percent of GDP at end-2006 to half this level in 2012. Similarly, the debt service-to-revenue ratio decreases from about 78 percent of the government revenues in 2006 to 31 percent in 2012. The sensitivity analysis indicates, however, that the pace of debt reduction may be significantly affected by policy reversal or exogenous shocks, which would increase the risk of debt distress over the medium term. In the no reform scenario, in which the primary balance is maintained at its 2006 level , debt burden indicators remain well above the baseline scenario, although they continue to improve over time. A one-time 30 percent real depreciation in the Guinean franc and real GDP growth at historical level minus one standard deviation in 2008–09 (the most extreme stress tests) would also generate a significant increase in the debt burden indicators (Figure 5).

D. Conclusion

14. **The debt sustainability analysis concludes that under the baseline scenario Guinea is and would remain at a high risk of debt distress into the medium term** in the absence of the country's benefiting from the Enhanced HIPC Initiative and the Multilateral Debt Reduction Initiative (MDRI). Under the alternative scenario assuming that the HIPC completion point is reached by end-2008, the external debt burden would decline to a sustainable level from that point. However, even after benefiting from HIPC and MDRI, Guinea's debt sustainability would remain vulnerable to adverse exogenous shocks and policy reversals. These conclusions highlight the importance of a rapid attainment of the HIPC Completion Point, sustained implementation of sound macroeconomic and structural policies, and a prudent debt strategy. Results of the public sector DSA indicate that Guinea's domestic debt is significant but is expected to decrease over time. The sustainability analysis indicates, however, that the pace of public debt reduction may be significantly affected by policy reversal or exogenous shocks.

Figure 4. Guinea: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2007–27



Source: IMF and World Bank staffs projections and simulations.

[1] Concessionality of new borrowing and debt accumulation under the baseline scenario.

Table 2. Guinea: External Debt Sustainability Framework, Baseline Scenario, 2007–27 [1]
(Percent of GDP, unless otherwise indicated)

	Actual	Historical Average [7]	Standard Deviation [7]	Projections									
	2006			2007	2008	2009	2010	2011	2012	2007–12 Average	2017	2027	2013–27 Average
External debt (nominal) [1]	**101.5**			**65.7**	**67.8**	**65.5**	**62.6**	**59.7**	**57.1**		**42.9**	**22.5**	
Of which: public and publicly guaranteed (PPG)	101.5			65.7	67.8	65.5	62.6	59.7	57.1		42.9	22.5	
Change in external debt	3.4			-35.8	2.1	-2.3	-2.9	-2.9	-2.5		-2.6	-1.7	
Identified net debt-creating flows	4.5			-0.1	-0.5	-2.9	-2.3	-1.6	-1.3		-1.2	-1.6	
Non-interest current account deficit	**4.5**	**4.1**	**2.5**	**8.4**	**12.4**	**10.3**	**11.7**	**14.3**	**14.3**		**13.6**	**12.3**	13.3
Deficit in balance of goods and services	4.6			8.8	12.1	11.1	12.5	15.0	14.9		14.0	12.4	
Exports	34.9			25.6	27.5	28.2	28.0	27.1	27.1		26.4	24.8	
Imports	39.6			34.5	39.5	39.4	40.5	42.1	42.0		40.4	37.2	
Net current transfers (negative = inflow)	-0.1	-0.2	0.6	-0.2	0.1	0.1	0.0	0.0	0.0		0.1	0.1	0.1
Of which: official	-0.6			-0.5	-0.3	-0.1	-0.1	-0.1	-0.1		-0.1	-0.1	
Other current account flows (negative = net inflow)	-0.1			-0.2	0.2	-0.9	-0.8	-0.7	-0.7		-0.5	-0.2	
Net FDI (negative = inflow)	**-5.1**	**-1.3**	**2.0**	**-8.5**	**-10.1**	**-10.7**	**-11.5**	**-13.3**	**-13.3**		**-13.2**	**-13.1**	-13.2
Endogenous debt dynamics [2]	**5.1**			**0.0**	**-2.8**	**-2.4**	**-2.6**	**-2.6**	**-2.3**		**-1.6**	**-0.8**	
Contribution from nominal interest rate	1.4			1.0	0.5	0.9	0.8	0.7	0.7		0.5	0.3	
Contribution from real GDP growth	-2.2			-1.1	-3.2	-3.3	-3.4	-3.3	-3.0		-2.1	-1.1	
Contribution from price and exchange rate changes[3]	6.0			-30.6	…	…	…	…	…		…	…	
Residual (3–4) [4]	**-1.1**			**-5.1**	**2.6**	**0.6**	**-0.6**	**-1.3**	**-1.2**		**-1.5**	**-0.1**	
Of which: exceptional financing	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt [5]	64.9			41.0	43.1	41.2	39.1	37.6	36.3		28.6	16.0	
Percent of exports	185.7			159.8	156.9	145.9	139.9	138.6	134.1		108.5	64.4	
NPV of PPG external debt	**64.9**			**41.0**	**43.1**	**41.2**	**39.1**	**37.6**	**36.3**		**28.6**	**16.0**	
Percent of exports	**185.7**			**159.8**	**156.9**	**145.9**	**139.9**	**138.6**	**134.1**		**108.5**	**64.4**	
Percent of government revenues	**530.0**			**363.7**	**313.4**	**282.1**	**255.0**	**244.8**	**233.3**		**185.2**	**103.3**	
Debt service-to-exports ratio (percent)	**18.6**			**16.4**	**7.6**	**12.7**	**12.2**	**10.3**	**9.7**		**8.1**	**6.0**	
PPG debt service-to-exports ratio (percent)	**18.6**			**16.4**	**7.6**	**12.7**	**12.2**	**10.3**	**9.7**		**8.1**	**6.0**	
PPG debt service-to-revenue ratio (percent)	**53.2**			**37.3**	**15.2**	**24.5**	**22.2**	**18.2**	**16.9**		**13.9**	**9.7**	
Total gross financing need (billions of U.S. dollars)	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.1	
Noninterest current account deficit that stabilizes debt ratio	1.1			44.2	10.3	12.6	14.6	17.2	16.8		16.3	14.0	
Key macroeconomic assumptions													
Real GDP growth (percent)	2.2	3.4	1.2	1.5	4.9	5.2	5.5	5.7	5.4	4.7	5.0	5.0	5.0
GDP deflator in U.S. dollar terms (change in percent)	-5.7	-4.9	9.1	43.2	-5.4	0.8	1.3	2.1	2.3	7.4	3.0	3.4	3.2
Effective interest rate (percent) [6]	1.4	0.9	0.7	1.5	0.7	1.4	1.3	1.2	1.2	1.2	1.2	1.5	1.3
Growth of exports of G&S (U.S. dollar terms, percent)	18.7	4.2	7.1	6.6	6.3	9.2	5.7	4.7	7.7	6.7	7.2	7.6	7.8
Growth of imports of G&S (U.S. dollar terms, percent)	21.1	3.2	9.3	26.6	13.7	5.7	9.9	12.2	7.6	12.6	7.3	7.6	7.6
Grant element of new public sector borrowing (percent)	…	…	…	43.4	40.0	39.3	39.3	38.1	37.7	39.6	36.0	35.1	35.8
Aid flows (millions of U.S. dollars) [8]	109.1			146.0	219.4	195.2	228.4	209.4	216.6		219.8	231.7	
Of which: Grants	39.9			79.4	71.5	66.9	70.9	72.6	73.1		76.3	88.2	
Grant-equivalent financing (percent of GDP) [9]	…			2.3	2.9	3.4	3.2	2.7	2.6		1.7	0.9	1.5
Grant-equivalent financing (percent of external financing) [9]	…			75.6	59.7	52.2	53.0	54.1	52.9		52.5	50.8	51.5
Memorandum items:													
Nominal GDP (billions of U.S. dollars)[3]	3.1			4.6	4.5	4.8	5.1	5.5	6.0		9.0	20.1	
(NPVt-NPVt-1)/GDPt-1 (percent)				-5.4	1.8	0.6	0.6	1.5	1.6	0.1	0.9	0.2	0.8

Source: IMF and World Bank Staffs simulations.

[1] Includes both public and private sector external debt.
[2] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
[3] Strong appreciation of the local currency in 2007 boosted the nominal GDP in U.S. dollar terms, which greatly contributed to the reduction of external debt to GDP ratio.
[4] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
[5] Assumes that NPV of private sector debt is equivalent to its face value.
[6] Current-year interest payments due divided by previous period debt stock.
[7] Historical averages and standard deviations are derived over the past 10 years.
[8] Defined as grants, concessional loans, and debt relief.
[9] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Table 3. Guinea: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007–27
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of debt-to-GDP ratio								
Baseline	41	43	41	39	38	36	**29**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 [1]	41	44	48	51	55	59	**82**	136
A2. New public sector loans on less favorable terms in 2008–27 [2]	41	44	43	42	41	40	**34**	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	41	44	44	41	40	38	**30**	17
B2. Export value growth at historical average minus one standard deviation in 2008–09 [3]	41	45	46	44	42	41	**32**	17
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008–09	41	47	53	50	49	47	**37**	21
B4. Net nondebt creating flows at historical average minus one standard deviation in 2008–09 [4]	41	50	55	53	51	49	**37**	18
B5. Combination of B1–B4 using one-half standard deviation shocks	41	47	54	51	49	48	**37**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	41	63	60	57	55	53	**42**	23
NPV of debt-to-exports ratio								
Baseline	160	157	146	140	139	134	**108**	64
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 [1]	160	160	168	181	202	218	**311**	549
A2. New public sector loans on less favorable terms in 2007–26 [2]	160	159	151	149	150	148	**130**	92
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	160	157	146	140	139	134	**108**	64
B2. Export value growth at historical average minus one standard deviation in 2008–09 [3]	160	178	200	193	191	185	**147**	83
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	160	157	146	140	139	134	**108**	64
B4. Net nondebt creating flows at historical average minus one standard deviation in 2008–09 [4]	160	182	196	189	188	181	**140**	74
B5. Combination of B1–B4 using one-half standard deviation shocks	160	169	180	173	171	166	**133**	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	160	157	146	140	139	134	**108**	64
NPV of debt-to-revenue ratio								
Baseline	364	313	282	255	245	233	**185**	103
A. Alternative Scenarios								
A1. Key variables at their historical averages in 200–26 [1]	364	320	325	331	356	378	**531**	880
A2. New public sector loans on less favorable terms in 2007–26 [2]	364	318	293	271	265	257	**222**	148
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	364	322	298	269	259	246	**196**	109
B2. Export value growth at historical average minus one standard deviation in 2008–09 [3]	364	324	315	285	274	261	**204**	108
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008–09	364	345	364	329	316	301	**239**	133
B4. Net nondebt creating flows at historical average minus one standard deviation in 2008–09 [4]	364	364	379	345	331	316	**239**	118
B5. Combination of B1–B4 using one-half standard deviation shocks	364	341	369	334	321	306	**240**	130
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	364	459	413	373	358	342	**271**	151

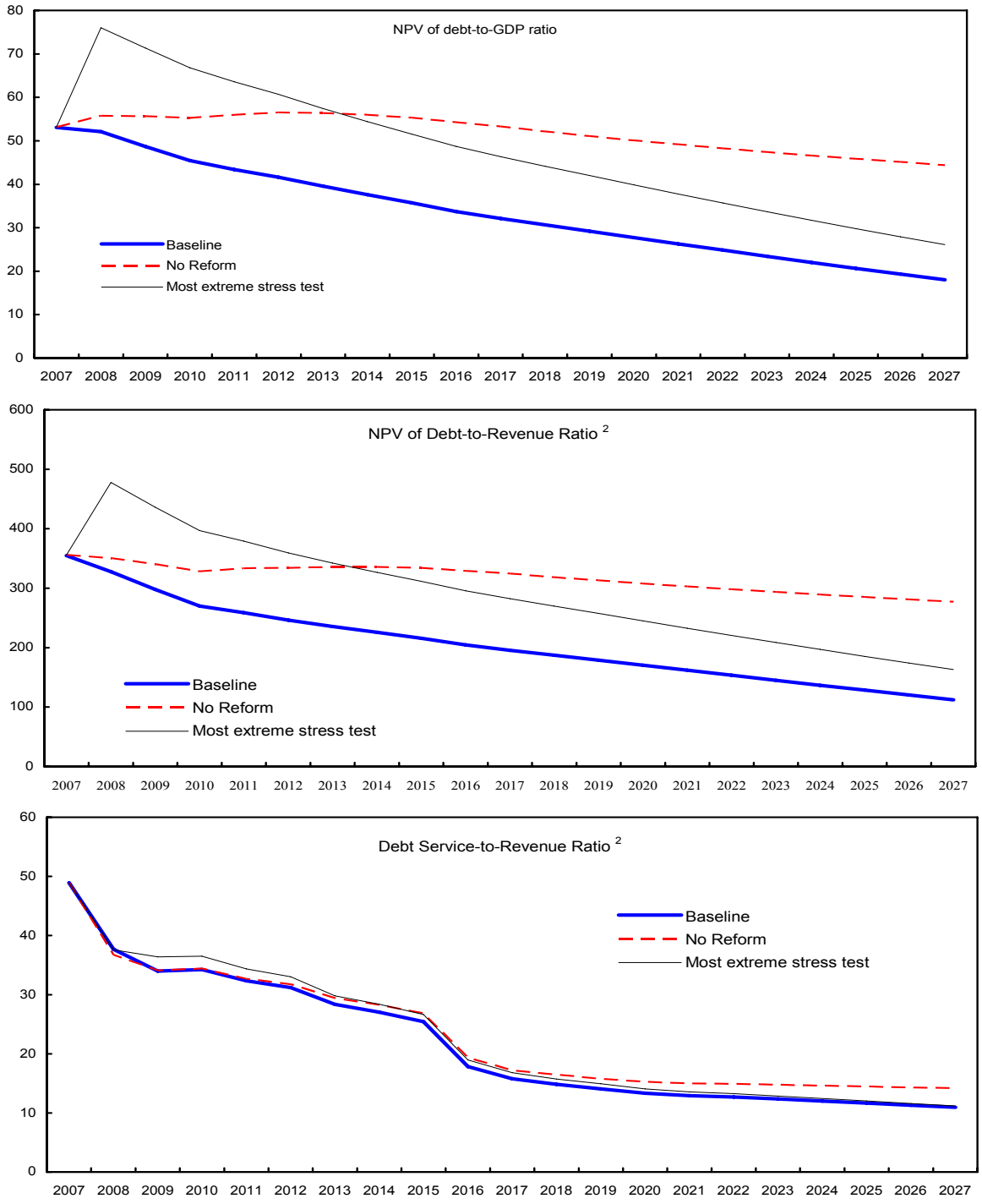
Table 3. Guinea: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007–27 (concluded)
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
Debt service-to-exports ratio								
Baseline	16	8	13	12	10	10	**8**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 [1]	16	8	14	15	14	14	**22**	47
A2. New public sector loans on less favorable terms in 2008–27 [2]	16	8	13	13	11	10	**9**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	16	8	13	12	10	10	**8**	6
B2. Export value growth at historical average minus one standard deviation in 2008–09[3]	16	8	16	15	13	12	**11**	8
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008–09	16	8	13	12	10	10	**8**	6
B4. Net nondebt creating flows at historical average minus one standard deviation in 2008–09 [4]	16	8	13	13	11	11	**11**	7
B5. Combination of B1–B4 using one-half standard deviation shocks	16	8	15	14	12	11	**10**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008[5]	16	8	13	12	10	10	**8**	6
Debt service-to-revenue ratio								
Baseline	37	15	24	22	18	17	**14**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27[1]	37	15	27	27	24	25	**37**	76
A2. New public sector loans on less favorable terms in 2008–27[2]	37	15	25	23	20	18	**15**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	37	16	26	23	19	18	**15**	10
B2. Export value growth at historical average minus one standard deviation in 2008–09[3]	37	15	25	23	19	17	**16**	10
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008–09	37	17	32	29	24	22	**18**	12
B4. Net nondebt creating flows at historical average minus one standard deviation in 2008–09 [4]	37	15	25	24	20	19	**19**	12
B5. Combination of B1–B4 using one-half standard deviation shocks	37	16	30	27	23	21	**18**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008[5]	37	22	36	32	27	25	**20**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline)[6]	38	38	38	38	38	38	**38**	38

Source: IMF and World Bank staffs projections and simulations.

[1] Variables include real GDP growth, growth of GDP deflator (U.S. dollar terms), non-interest current account in percent of GDP, and nondebt creating flows.
[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
[4] Includes official and private transfers and FDI.
[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 5. Guinea: Indicators of Public Debt Under Alternative Scenarios, 2007–27 [1]



Source: IMF and World Bank staffs projections and simulations.

[1] Most extreme stress test is test that yields highest ratio in 2017.

[2] Revenue including grants.

Table 4. Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2006-2027
(Percent of GDP, unless otherwise indicated)

	Actual			Projections									
	2006	Historical Average [5]	Standard Deviation [5]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt [1]	116.8			77.8	76.9	73.0	68.9	65.5	62.4		46.3	24.3	
Of which: external debt	101.5			65.7	67.8	65.5	62.6	59.7	57.1		42.9	22.5	
Change in public sector debt	-16.6			-39.0	-1.0	-3.9	-4.1	-3.5	-3.0		-2.8	-1.8	
Identified debt-creating flows	-23.5			-37.0	-2.0	-4.6	-4.7	-5.3	-4.8		-3.2	-2.3	
Primary deficit	-0.4	1.4	1.7	-2.4	-1.1	-1.2	-1.2	-1.4	-1.2	-1.4	-0.3	-0.7	-0.5
Revenue and grants	14.7			14.9	15.9	16.3	16.9	16.8	17.0		16.5	16.1	
Of which: grants	1.4			2.0	1.6	1.4	1.4	1.3	1.2		0.9	0.4	
Primary (noninterest) expenditure	14.3			12.6	14.8	15.1	15.7	15.4	15.7		16.2	15.4	
Automatic debt dynamics	-19.2			-32.4	1.0	-3.3	-3.5	-3.8	-3.6		-2.9	-1.6	
Contribution from interest rate/growth differential	-5.7			-4.6	-3.9	-4.0	-3.9	-3.7	-3.4		-2.5	-1.3	
Of which : contribution from average real interest rate	-2.9			-2.8	-0.2	-0.2	-0.1	0.0	0.0		-0.1	0.0	
Of which : contribution from real GDP growth	-2.8			-1.7	-3.6	-3.8	-3.8	-3.7	-3.3		-2.3	-1.3	
Contribution from real exchange rate depreciation	-13.5			-27.9	4.8	0.7	0.4	-0.1	-0.2		...	...	
Other identified debt-creating flows	-3.9			-2.2	-1.9	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-1.7			-0.9	-0.5	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-2.2			-1.3	-1.4	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	6.9			-2.0	1.1	0.7	0.6	1.8	1.8		0.4	0.4	
NPV of public sector debt	80.2			53.0	52.1	48.7	45.4	43.4	41.6		32.0	17.8	
Of which: external debt	64.9			41.0	43.1	41.2	39.1	37.6	36.3		28.6	16.0	
NPV of contingent liabilities (not included in public sector debt)	...			...	...	...	...	...	...		...	...	
Gross financing need [2]	12.5			6.9	6.5	5.7	6.0	5.4	5.3		3.2	1.5	
NPV of public sector debt-to-revenue ratio (percent) [3]	546.1			355.0	327.5	298.6	269.1	257.6	245.3		194.3	111.0	
Of which: external	441.9			274.1	270.7	252.6	231.6	223.4	214.2		173.8	99.4	
Debt service-to-revenue ratio (percent) [3, 4]	78.2			48.9	37.7	34.1	34.1	32.3	31.1		15.7	10.9	
Primary deficit that stabilizes the debt-to-GDP ratio	16.2			36.6	-0.2	2.6	2.9	2.1	1.8		2.5	1.1	
Key macroeconomic and fiscal assumptions													
Real GDP growth (percent)	2.2	3.4	1.2	1.5	4.9	5.2	5.5	5.7	5.4	4.7	5.0	5.0	5.0
Average nominal interest rate on forex debt (percent)	1.9	1.5	0.3	1.2	0.8	1.5	1.3	1.2	1.2	1.2	1.2	1.5	1.3
Average real interest rate on domestic currency debt (percent)	-12.2	-1.7	7.0	-9.2	4.4	0.8	4.0	6.1	6.6	2.1	5.4	6.1	4.6
Real exchange rate depreciation (percent, + indicates depreciation)	-11.5	8.6	16.6	-28.3	...	...	...	...	...	...	...	...	...
GDP deflator (percent)	34.7	12.1	11.9	19.8	13.1	6.8	4.3	4.1	4.4	8.7	5.1	5.5	5.3
Growth of real primary spending (deflated by GDP deflator, percent)	17.3	3.6	12.9	-10.8	23.6	7.1	9.9	3.8	7.3	6.8	6.4	4.4	4.9

Sources: Guinean authorities; and IMF and World Bank staffs estimates and projections.

[1] Public sector debt covers only central government debt.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues including grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are derived over the past 10 years.

Table 5. Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2007-2027

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	53	52	49	45	43	42	32	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	53	55	55	55	56	56	53	44
A2. Primary balance is unchanged from 2006	53	56	56	55	56	57	53	44
A3. Permanently lower GDP growth [1]	53	52	49	46	45	44	36	24
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	53	54	53	50	49	48	40	27
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	53	56	56	52	50	48	36	20
B3. Combination of B1-B2 using one half standard deviation shocks	53	56	57	53	51	48	37	20
B4. One-time 30 percent real depreciation in 2008	53	76	71	67	64	61	46	26
B5. 10 percent of GDP increase in other debt-creating flows in 2008	53	62	57	53	50	48	37	20
NPV of Debt-to-Revenue Ratio [2]								
Baseline	355	328	298	270	259	246	196	112
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	356	347	336	324	330	332	322	271
A2. Primary balance is unchanged from 2006	356	350	341	328	334	334	324	277
A3. Permanently lower GDP growth [1]	356	330	302	276	267	257	216	150
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	356	338	322	297	290	281	242	165
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	356	353	345	311	297	281	221	125
B3. Combination of B1–B2 using one half standard deviation shocks	356	354	349	315	301	286	225	127
B4. One-time 30 percent real depreciation in 2008	356	478	436	397	379	359	282	163
B5. 10 percent of GDP increase in other debt-creating flows in 2008	356	387	349	315	300	284	223	126
Debt Service-to-Revenue Ratio [2]								
Baseline	49	38	34	34	32	31	16	11
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	49	37	35	36	35	34	19	16
A2. Primary balance is unchanged from 2006	49	37	34	34	33	32	17	14
A3. Permanently lower GDP growth [1]	49	37	34	34	32	31	15	11
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	49	38	35	36	34	33	16	11
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	49	37	34	34	32	31	15	10
B3. Combination of B1–B2 using one half standard deviation shocks	49	37	35	36	34	32	16	10
B4. One-time 30 percent real depreciation in 2008	49	38	36	36	34	33	17	11
B5. 10 percent of GDP increase in other debt-creating flows in 2008	49	37	35	35	33	31	15	10

Sources: Guinea authorities; and IMF and World Bank staffs estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

[2] Revenues are defined inclusive of grants.